

- 4 The photograph shows two views of an organelle, observed using an electron microscope.



organelle

(Source: © Biology Pics / Science Photo Library)

- (a) (i) Which is the correct function of this organelle?

(1)

- ☐ A formation of lysosomes
- ☐ B formation of microtubules
- ☐ C formation of ribosomal RNA
- ☐ D formation of vesicles

- (ii) Suggest why the two views of this organelle look different.

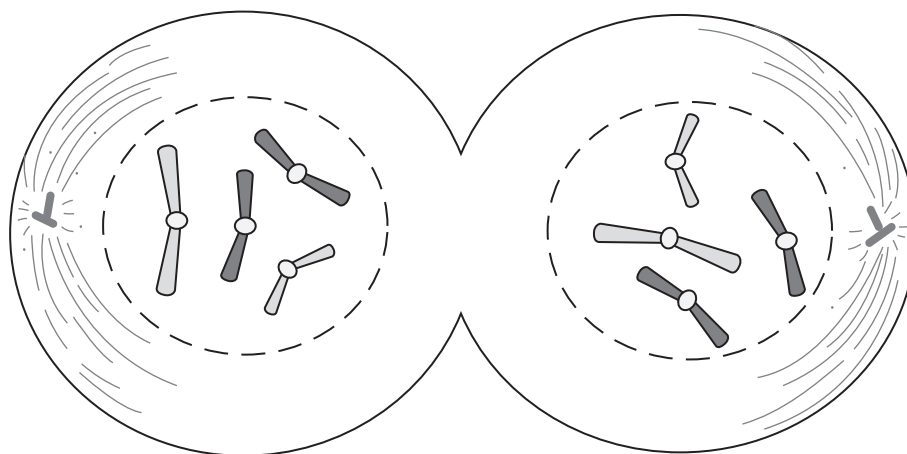
(1)

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(b) The diagram shows a cell in telophase of mitosis.



(Source adapted from: <https://www.shutterstock.com/image-vector/telophase-phase-cell-cycle-1678454008>)

(i) Draw **this** cell to show it in **metaphase** of mitosis.

(3)

(ii) Describe what happens in **anaphase** of mitosis.

(2)

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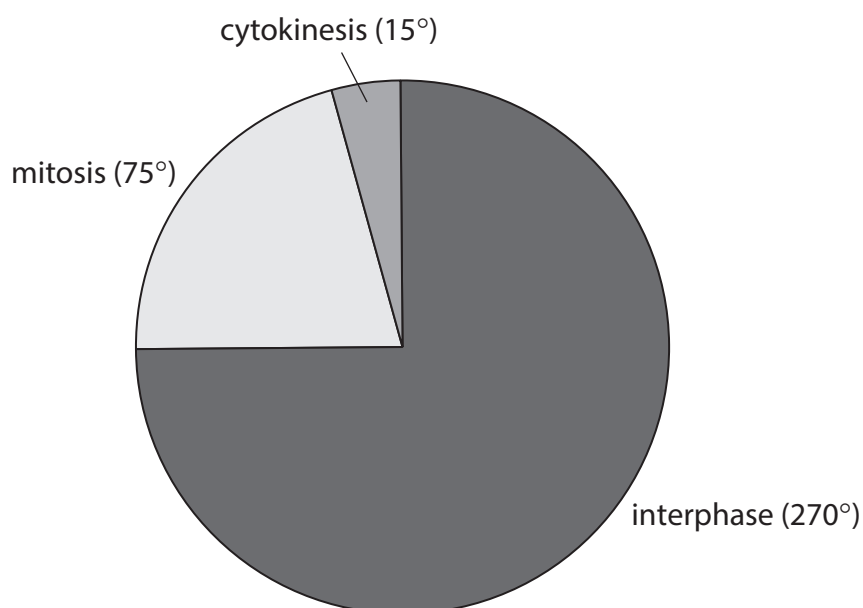
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- (c) The diagram shows the relative proportions of the time spent in each stage of the cell cycle.

In these cells, cytokinesis takes 20 minutes.



Calculate how long interphase would take in these cells.

Give your answer in **hours**.

(1)

..... hours

- (d) State how the mitotic index of a tissue could be calculated.

(1)

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(Total for Question 4 = 9 marks)